Table 9: Sensitivity of failure-mode counts to the sign tolerance ϵ. RH denotes reward hacking; SA stable alignment; OC optimization collapse; PUA proxy under-alignment; CS conservative stagnation; MA mixed/ambiguous; EG evaluator gaming.

Unit	ϵ	Total	RH	SA	OC	PUA	CS	MA	EG
Checkpoint	10^{-8}	31	4	7	5	6	0	9	14
Checkpoint	10^{-3}	31	4	7	5	6	0	9	14
Checkpoint	10^{-2}	31	4	7	5	5	1	9	13
Checkpoint	5×10^{-2}	31	3	4	1	1	6	16	5
Checkpoint	10^{-1}	31	0	1	1	0	18	11	1
Row-level	10^{-8}	1920	127	100	99	122	748	724	75
Row-level	10^{-3}	1920	127	98	99	122	749	725	75
Row-level	10^{-2}	1920	123	96	99	120	762	720	75
Row-level	5×10^{-2}	1920	115	83	92	110	823	697	75
Row-level	10^{-1}	1920	105	72	79	101	867	696	75

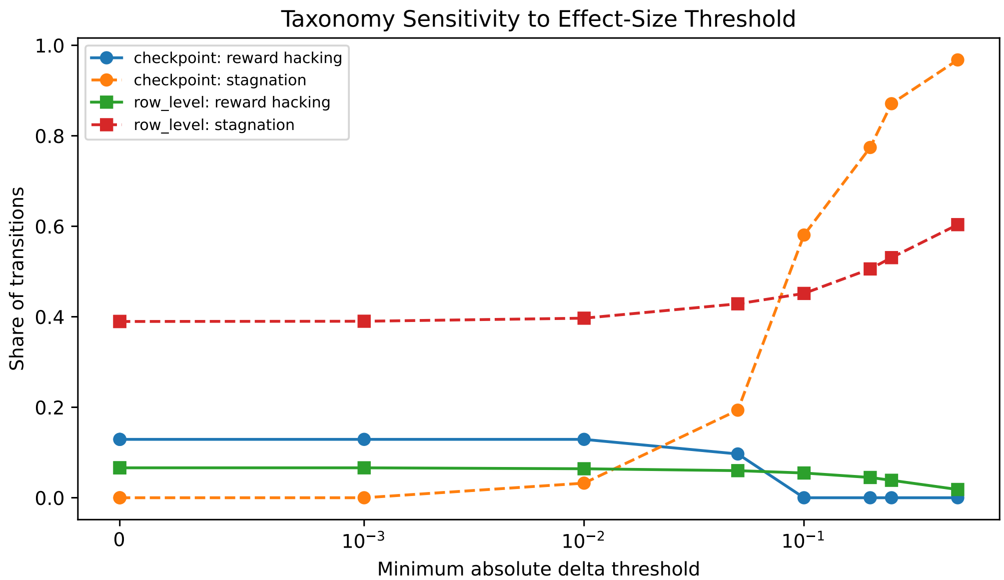

Figure 9: Sensitivity of taxonomy counts to minimum absolute delta thresholds. Larger thresholds make the classifier more conservative, shifting small movements into conservative stagnation or mixed/ambiguous classes.

mode. The distinction between checkpoint and row-level disagreement is important. At checkpoint level, 14 of 31 transitions exhibit evaluator gaming, producing a high aggregate disagreement rate. At row level, only 75 of 1,920 transitions exhibit the same opposite-direction pattern. This suggests that checkpoint-level judge disagreement can be driven by small average shifts across many prompts, while row-level disagreement is more localized.

At row level, proxy under-alignment has the highest evaluator-gaming share (17.2%), followed by stable alignment (15.0%) and reward hacking (11.8%). Conservative stagnation has zero evaluator-gaming events by construction in this artifact set because both the proxy and primary judge are approximately unchanged. The higher disagreement rates in proxy under-alignment and reward hacking are consistent with the central diagnosis of the paper: when the proxy and evaluator disagree, the two evaluators are also more likely to expose instability in what is being measured.

These results motivate keeping $R^\dagger$, $R_2^\dagger$, and $\bar{R}^\dagger$ visible as separate quantities. Averaging judges can stabilize noisy evaluations, but it can also hide evaluator-specific movement. For failure analysis, the

Table 10: Representative checkpoint transitions where aggregate and row-level diagnoses diverge. CK denotes checkpoint label; Dom. denotes dominant row label; PUA proxy under-alignment; SA stable alignment; OC optimization collapse; MA mixed/ambiguous.

Setting	Step	CK	Dom.	Row RH share	Dom. share
Aggressive PPO	$600 \rightarrow 1000$	PUA	MA	0.188	0.438
$\lambda = 0.5$ UP-PPO	$600 \rightarrow 1000$	PUA	MA	0.141	0.609
$\beta = 0.001$ sampled PPO	$200 \rightarrow 500$	SA	MA	0.141	0.438
Aggressive PPO	$200 \rightarrow 400$	PUA	MA	0.109	0.594
Aggressive PPO	$400 \rightarrow 600$	SA	MA	0.109	0.500
$\beta = 0.03$ PPO	$250 \rightarrow 350$	OC	MA	0.109	0.406

Table 11: Evaluator-gaming share by failure mode. Judge gap is $|\Delta R^{\dagger} - \Delta R_2^{\dagger}|$.

Unit	Failure mode	Total	Eval. gaming	Mean gap	Median gap	Share
Row	Proxy under-alignment	122	21	1.516	1.000	0.172
Row	Stable alignment	100	15	1.275	1.000	0.150
Row	Reward hacking	127	15	1.138	1.000	0.118
Row	Optimization collapse	99	5	0.904	1.000	0.051
Row	Mixed/ambiguous	724	19	0.800	1.000	0.026
Row	Conservative stagnation	748	0	0.167	0.000	0.000
Checkpoint	Optimization collapse	5	3	0.242	0.180	0.600
Checkpoint	Stable alignment	7	4	0.206	0.164	0.571
Checkpoint	Mixed/ambiguous	9	4	0.043	0.039	0.444
Checkpoint	Proxy under-alignment	6	2	0.118	0.094	0.333
Checkpoint	Reward hacking	4	1	0.166	0.145	0.250

disagreement event is itself informative: it marks transitions where the measured direction of progress depends on the evaluator used to define progress.

We also checked whether the stored row-level length features suggest a simple verbosity explanation. Reward-hacking rows have higher target length than non-reward-hacking rows on average (59.3 versus 52.4 words), but they do not become longer across the transition: mean length change is -3.88 words for reward-hacking rows and -2.68 words for other rows. The correlation between target length and reward-hacking status is small ($r = 0.083$), as is the correlation between length change and reward-hacking status ($r = -0.018$). These diagnostics do not replace length-controlled judging, but they make a pure verbosity account less plausible in the current artifacts.

H Code and Artifact Availability

The repository accompanying this paper contains the paper source, generated tables, figures, judge prompts, parsing logic, and reproducibility artifacts needed to inspect the reported analyses. Because external judge versions and API behavior can change over time, the stored score artifacts are part of the replication record. The repository is available at: https://github.com/zabahana/rlhf-failure-modes-diagnostics. A live interactive demo of the model comparator and diagnostic views is available at: https://rlhf-failures.zelalem.ai/.